Measurement Center Processes Support in the Automotive Industry

Conference Paper *in* Advances in Intelligent Systems and Computing · September 2018

DOI: 10.1007/978-3-319-67223-6_28

Martin Sarnovsky
Technical University of Košice

Measurement Center Processes Support in the Automotive Industry

Martin Sarnovsky(✉) and Petra Cibulova

Department of Cybernetics and Artificial Intelligence,
Faculty of Electrotechnics and Informatics,
Technical University Kosice, Letna 9/A, 04001 Kosice, Slovakia
{martin.sarnovsky, petra.cibulova}@tuke.sk

Abstract. Nowadays, business process management and process modeling is an important part of business management and business process improvement. Resulting from the increasingly fast growing customer requirements for the final product, organizations need to introduce process management. The work presented in this paper was situated in the area of the automotive industry, which is the most important sector of the world economy. Technical development is progressing with rapid fashion, and even in the automotive industry, there is a need to monitor, manage and improve production processes. This is related to the deployment and use of information systems that support and automate these processes. The main goal of the work described in the paper is to provide a brief overview of the field of business process management, process modeling, related technologies, and the situation of the automotive industry in general. This work also analyzes the current situation in one of the measurement centers, selected processes, and user requirements for the application supporting those processes. Based on the analysis, we proposed an improvement of the selected processes and also designed a web application that supports these processes. The application was built using current technologies such as ASP.NET Core Framework, MVC architecture and the available libraries. Results of testing proved that the solution meets all requirements and in terms of key performance indicators surpassed the expected results. It is suitable for real use in practice and in the future, it is recommended its deployment to all measurement centers.

Keywords: Business processes · Process modeling · Process optimization · Automotive industry · Measurement center

1 Introduction

Nowadays, the automotive industry is one of the most important sectors of the world economy and even of the economy of the Slovak Republic. Slovakia is currently one of the top automotive manufacturers in the world. Measurement centers play an important role in the whole production process. The main objective of the measurement centers is to detect the smallest variations between the real parameters of the manufactured automotive parts and the required ones, thus preventing the errors of the final vehicles. The work presented in this paper was performed in the environment of one particular measurement center in chosen car factory. The work is dedicated to a thorough analysis

Z. Wilimowska et al. (eds.), *Information Systems Architecture and Technology: Proceedings of 38th International Conference on Information Systems Architecture and Technology – ISAT 2017*, Advances in Intelligent Systems and Computing 657, DOI 10.1007/978-3-319-67223-6_28

of the current situation at a specific measurement center and based on the analysis we propose the most suitable solution for improvement of the selected processes deployed at this measurement center.

2 Automotive Industry and Manufacturing Process

In general, automotive industry involves a cooperation of a large number of companies and different organizations in various stages of the production of the motor vehicles such as design, development, production, marketing, and sales [1]. It is one of the largest and most important sectors of the world economy, especially in Slovakia. Nowadays Slovakia belongs to the 20 biggest car producers in the World. In Slovakia, there are 3 big world car manufacturers, and since 2018 one more will start the production there [2]. According to production statistics [3], since 2015, Slovakia has reached 1 million vehicles produced per year. Taking into consideration the number of inhabitants, Slovakia is by far the global leader in car production per 1000 inhabitants [2]. Figure 1 depicts the number of cars produced in Slovakia per year with the forecast for the year 2020 when the estimated number of cars produced in the country will reach 1 350 000 units per year.

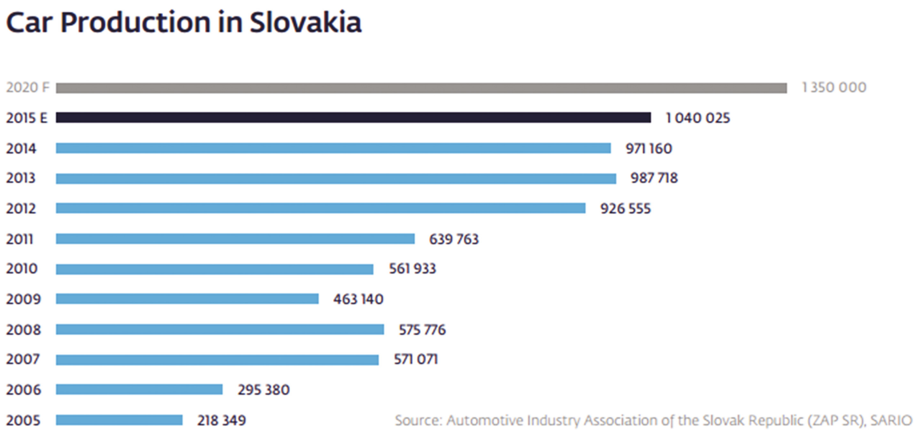

Fig. 1. Car production in the Slovak Republic

In general, the car manufacturing process can be described as follows: everything starts with the metal sheets and the handling process that takes place in the press shop. The sheets are cut to the required dimensions, then pressed and bent. The first moldings, after each mold change, are scanned measured at the measurement centers. From these moldings, the car body is then welded and goes to the paint shop, where all its parts are thoroughly painted with the desired color. In this final car body, there are cable networks located and the dashboard is put together. Then, there are put together all necessary car parts (such as engine, drive mechanism, wheels, etc.). This is where the production process ends and the car passes from the production hall to the test track where the car goes through the testing. After success in the testing phase, the car can be shipped to the customer.

3 Current Situation at Measurement Center

All manufactured car parts must pass through the Measurement center. Only those that meet all the necessary standards and strict criteria can be returned to the production process. The measurement center for which the output of this work is designed and developed is one of the three measurement centers that are located in the selected automotive production. This particular measurement center includes four touch measuring machines (KMG). Currently, they are delivered, measured, and analyzed on an average of 500 pieces per month. The current process of the transport of the measured vehicle parts is depicted in Fig. 2.

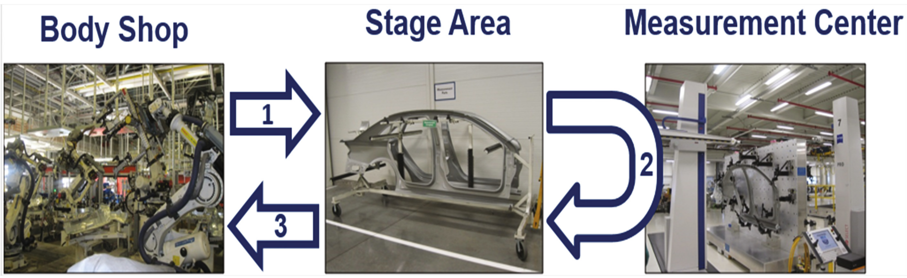

Fig. 2. Transport process of the measured parts

According to the pre-established measurement plan, the production workers move the necessary parts to the Stage area. The Stage area is a place that temporarily stores the parts that need to be measured or that have already passed the measurement process. From this place, the workers of the measurement center move the necessary parts to the measuring machines, where the particular measurements and analyses will be carried out. Upon completion of the measurements, the parts that have successfully passed through will be moved back from the measuring center to the Stage area. Finally, production workers will pick up the measured parts from the Stage area and move them back to the Body Shop. This process is currently not formalized (in any way) and not supported by any software in the measurement center. The staff of individual centers is not sufficiently informed about the actual status of the parts at the measurement center, which often causes problems. One of the most important ones is that the Stage area is filled with the temporarily stored parts too fast. Because of the lack of space there, there is a need to expand this area. At present, 1,233.94 m^2 space is used for the Stage area. Measurement center managers estimate, that the expansion of the center would cost more than 1,200 EUR per square meter.

3.1 Current Problems in the Measurement Center

The main reason of the problems is the lack of information and communication between the Body Shop and the Measurement center. In this case, especially the obsolete, “paper” way of creating, approving, and distributing measurement plans is implemented. This method does not allow any real-time information exchange when operational changes apply to these plans. Plans themselves are currently being created in the Excel spreadsheet environment and, once created, they need to be approved by responsible workers, printed and distributed to the individual centers.

Because of the reasons mentioned above, it is necessary to create and deploy an appropriate information system and process support mechanism that will effectively support not only the transport of measured parts but also the creation, approval, and distribution of the measurement plans. Support for these processes should include the possibility of creating, approving and making available of measurement plans by several workers of different centers immediately. In addition, it should include the possibility of monitoring the status at the measurement center including production and logistics. It should enable the users to perform immediate notifications to all involved workers in case of a change in the measurement plan. Based on the information provided, it is possible for the production to effectively adapt to the changes. All this should be done in real-time and without unnecessary delays.

4 Proposed Solution

We proposed a solution, which automates certain stages of the current process and the workflows that take place at the Measurement center. The solution aims to optimize the existing processes [4] and to provide real-time communication between production, logistics and the Measurement center. Staff will exchange the necessary information through the proposed software solution supporting the designed and optimized processes. Such implementation will result in more readable generation of measurement plans on a daily, weekly and monthly basis.

4.1 Processes Re-engineering

Process of Transportation of the Parts to the Measurement Center

The process of the transportation of the parts to Measurement center was not formalized in any standard or notation. Therefore, we decided at first to design the process models and identify the concrete activities that could be supported by a kind of software solution in order to optimize the process or resources. The process is designed in a way, which enables the elimination of the issues mentioned in Sect. 3.1. The process was designed in the BPMN (Business Process Modeling Notation), which is a standard for modeling of the business processes [5] and is depicted in Fig. 3. The process starts when Measurement center staff receives the weekly measurements plan. The first step is to check, whether the actual state in the Measurement center is aligned with the plan. If not, the warning is issued and staff is notified. Notification has to be confirmed (staff has to confirm, that they are informed about the change). In the next step, the parts are ready for measurement and transported to the Stage area. Measurement center staff pick up the parts, update the information in the system, and move the part to perform measurements. When the part is located at measurement machine, staff changes the status of the part in the system and starts the measurement and analysis. The part remains in this state until the measurement in completed. After that, the part is moved back to the Stage area, its status is changed to “ready” and production staff is notified, that the part can be picked up from Stage area. Production staff confirms when the part is actually picked up from Stage area and moved back into the production. Figure 3 also shows the tasks directly supported by proposed software solution.

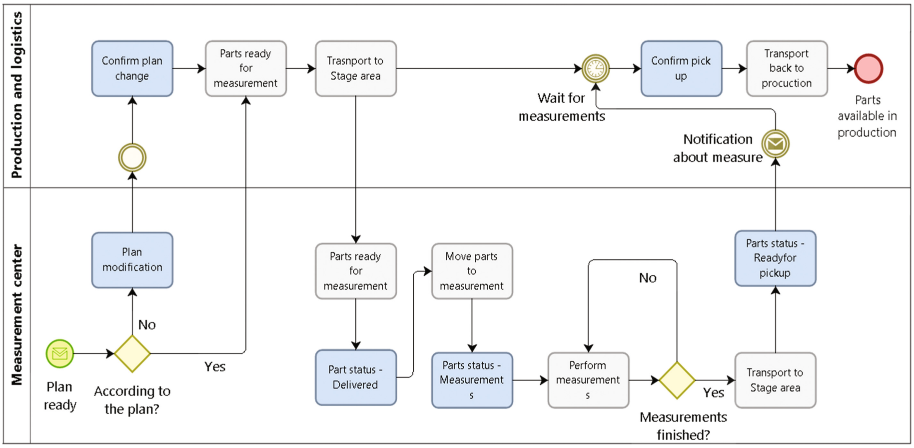

Fig. 3. Process of the transportation of the parts to the measurement center

Process of Measurements Planning

The planning phase is required to correctly perform the measurement process. In general, this process is performed on weekly basis and it results in the measurement plan which can be changed during the operation. The plan then orchestrates the all involved parties including Measurement center staff, Production staff, and Logistic staff. Changes during the operation can be made by the Measurement center designated staff members. Such operative changes should reflect the current requirements.

Designed and formalized process model is depicted in Fig. 4. Highlighted tasks are the ones supported and partially automatized by designed software solution. In general, the process starts weekly, when Measurement center staff member starts to create the plan, either from scratch or using the existing template. The template has to be checked, if it is ready to be used, in case when a new template is created, staff member has to fill all needed information (about the measuring machines and measurements) until all information for all machines is provided. When all information are provided and all fields correctly filled, designated worker sends the created plan for approval. In case of plan dismissal, designated worker performs the needed corrections (e.g. to remove errors) and again sends the plan for approval. When the plan is approved, it can be printed out or just stored in the database without printing. In both cases, the plan is available in the application for all involved parties.

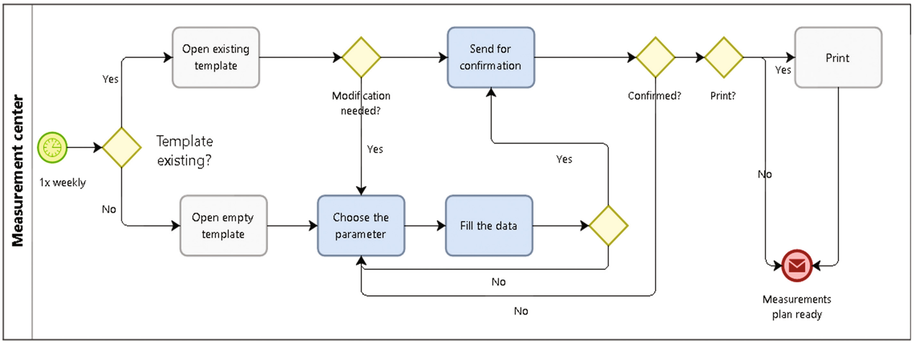

Fig. 4. Modified process of the measurement planning

4.2 Design and Implementation of the Supporting Application

The required functionalities of the proposed solution can be summed up:

- Definition of a measurement plan according to the process standard: possibility of defining a measurement plan for a particular measuring machine according to the type of measured parts. It is possible to enter the number of measured parts on a given measuring machine, to determine the time of measurement from - to, the possibility of adding, removing and adjusting the measurements.
- Approve and view the created plan.
- Possibility to make operational changes in an already generated and approved plan: adding a measurement, removing measurement and adjusting the measurement.
- All changes are immediately visible to all employees in real time: alerting workers to changes.
- Changing the status of the measured part: not delivered, delivered, done, measured, finished, picking up, transport to production.
- Search for possible measured parts.
- Application administration: setting permissions for individual application users, administration of the lists - vehicle model list, list of the types of measured parts, list of measuring machines.

After the initial analysis, we identified the three major user roles of the proposed system:

- Production and logistics staff.
- Measurement center staff.
- Administrators.

The main data model of the application is depicted in the Fig. 5. There are also more tables in the database that provide the functionalities of the user accounts management and roles support.

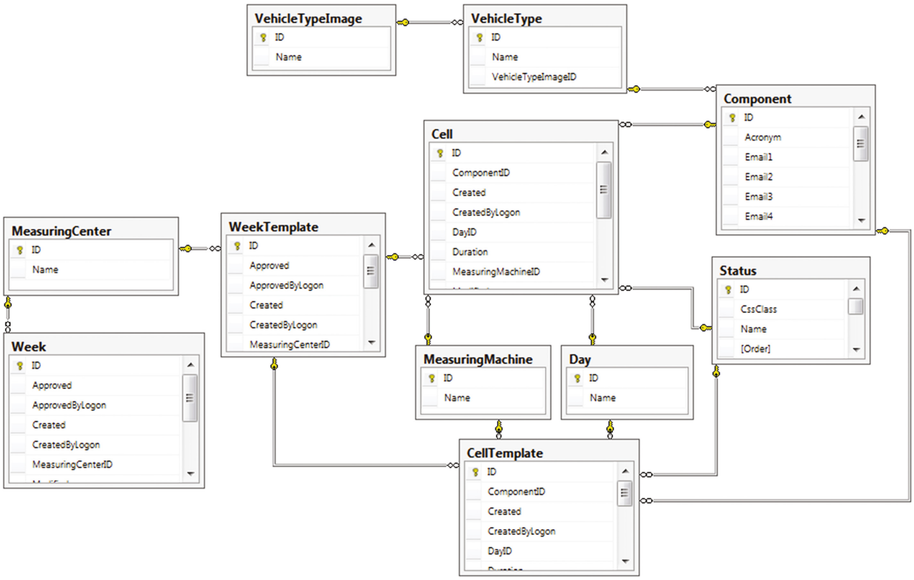

Fig. 5. Data model of the proposed solution

There are two main views that will provide the required functionality and one view for administrator that will manage lists. The first view is dedicated to the Production and Logistics staff as well as the staff of the Measurement center. All staff is able to view the current measurement plan for the particular day or week and set the status of individual measurements. This view also provides the alerts when the measurement plan has been changed.

The second view is dedicated for the staff of the Measurement center and provides the functionality to define and modify the measurement plan, search in existing records and lists. Users can create alerts displayed to the Production and Logistics staff at first view. We developed this application using Visual Studio 2017, the ASP.NET Core MVC framework [6] as the framework for web applications development. It leverages the MVC (Model-View-Controller) architecture [7]. We used Microsoft SQL Server 2014 database server. From programming languages, we used C#, HTML, CSS, ASP.NET, JavaScript and jQuery, DataTables and Bootstrap libraries.

5 Evaluation

To evaluate the proposed solution, we chose three key performance indicators (KPIs) [8]:

- The time needed to create, approve, and distribute the measurement plan (KPI I): currently, this time can climb to more than 3 h. We will consider the application to be successful if this time is reduced by at least 30%.
- The time in which the Stage area is filled in case of an operational change of the measurement plan (KPI II): when a delay occurs - at present if there is a delay of approximately 2 h, this space will be filled within one hour. We expect the application to be successful if this delay time is reduced by at least 30%, but it is best to avoid filling the Stage area.
- The time for which the Stage area is emptied in case of an operational change of the measurement plan (KPI III): at the present time, if the measurement is completed approximately 2 h earlier, this space begins to be emptied for up to two hours, as determined by the original plan. We will consider the application as successful if the Stage area is emptied within half an hour after the measured parts are delivered to this space.

6 Testing and Evaluation

Application testing took place in a simulated environment of the Measuring center for a period of three weeks. During the testing, the application was deployed to a test server where it was evaluated by chosen employees of the Measurement center. Server deployment, testing, and communications with the Measurement center has been provided by the company BSP Applications. The whole testing process took place in the form of Black-Box testing.

The first phase of the test trials lasted approximately two weeks. During this phase, we focused mostly on the detection of errors that would cause the application to crash, to specify the functionalities that would be missed in the real environment of the Measurement center by its end users.

The second phase of testing lasted approximately one week and this phase was mainly focused on tracking and estimating times for the predefined KPIs. User requirements were met, as the solution provided all the required functionalities and interfaces. KPI were evaluated by the staff of the Measurement center. Staff members were involved in a set of simulated processes and we measured the particular KPIs in several trial runs. The evaluation results proved that the KPI I was fulfilled as the expected time was peaking at 1 h 45 min, which represents the time reduction of 40% (30% planned). KPI II and KPI III metrics were tested in scenarios of the measurement delay of 2 h. During the simulations, the KPI II times were reduced by 40% and KPI III times by 70%.

7 Conclusion

The test results clearly indicate that the application met all user requirements and its measured performance based on predefined key performance indicators exceeds the expected results by at least 40% in each of these indicators. For this reason, we consider the solution to be successful and we can recommend that the solution is ready to be deployed to the real environment of the Measurement centers. In the future, it is possible to add the possibility of displaying images for individual types of cars in an aesthetic and design aspect. Also, it is recommended to extend the possibility of exporting weekly templates and plans to the SharePoint corporate environment as well as the possibility of sending SMS and emails with notifications in case of an operative change in the currently running weekly schedule.

Acknowledgements. The work presented in this paper was supported by the Slovak VEGA research grant 1/0493/16, Slovak APVV research grant APVV-16-0213 and Slovak KEGA grant 025TUKE-4/2015.

References

1. Rae, J.B., Binder, A.K.: Automotive Industry | Britannica.com. https://www.britannica.com/topic/automotive-industry
2. SARIO I Slovak Investment and Trade Development Agency: Automotive Sector in Slovakia (2016). http://www.sario.sk/sites/default/files/content/files/sario-automotive-sector-in-slovakia-2017-03-02.pdf
3. Production Statistics | OICA. http://www.oica.net/category/production-statistics/
4. Soković, M., Pavletić, D., Kern-Pipan, K.: Quality improvement methodologies - PDCA cycle, RADAR matrix, DMAIC and DFSS. J. Achiev. Mater. Manuf. Eng. **43**, 476–483 (2010)
5. White, S.A.: Introduction to BPMN (2004)
6. Freeman, A.: Pro ASP.NET Core MVC, 6th edn. Apress, New York (2016). ISBN 978-1-4842-0397-2
7. Galloway, J., et al.: Professional ASP.NET MVC 5. Wiley, Indianapolis (2014). ISBN 978-1-118-79475-3
8. Weber, A.: Maintenance Key Performance Indicators Introduction (2005). www.ivara.com
9. Indulska, M., et al.: Business process modeling: current issues and future challenges. In: The 21st International Conference on Advanced Information Systems, 8–12 June 2009, Amsterdam, The Netherlands (2009)
10. Recker, J., et al.: Business process modeling – a comparative analysis. J. Assoc. Inf. Syst. **10**(4), 333–363 (2009)